

SHARAD Design and Operation

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Abstract— This paper describes the operating principles and the design of the Mars Shallow Radar Sounder (SHARAD), an HF sounding radar devoted to the mapping of sub-surface features of Mars and currently operational on board the NASA/JPL's Mars Reconnaissance Orbiter spacecraft. Compared to its predecessor MARSIS, currently operating from ESA's Mars Express, SHARAD is characterised by an higher carrier frequency (20 MHz vs a max of 5 MHz for MARSIS) and a much wider signal bandwidth (10 MHz vs 1 MHz of MARSIS).

This allows SHARAD to achieve a finer range resolution (15 metres unweighted in free space) at the expenses of ground penetration, which makes the instrument ideal to probe the shallow subsurface layers (to depths of hundreds of meters) which cannot be resolved from the surface by the much far-reaching MARSIS.

SHARAD uses a 85 usec chirp signal with a PRF of 700 Hz and a peak power of 10 W, and radiates by means of a 10 meters Fiber Foldable Tube (FFT) dipole antenna, with a wide-band matching network in charge of impedance matching with the transceiver.

The most challenging requirement (especially considering the large fractional bandwidth of the system) is the level of the range sidelobes, which shall be below -55 dBc after the 6th lobe, to allow proper detection of the weak subsurface echoes in presence of strong surface returns. On this side, the design takes advantage from the wide download bandwidth made available by the MRO Spacecraft to keep the on-board processing to a minimum level (basically, only a programmable coherent presumming), and leave most of the processing (range compression and synthetic aperture) on ground. In this way it is easy to use Tx chirps and Rx transfer functions characterised on-ground as reference for range correlation, with the range sidelobes limited, basically, only by the stability of the RF hardware.

The limited amount of on-board processing also helped in limiting the complexity of the instrument design and, therefore, its mass and power consumption.

SHARAD uses a very simple architecture, with the transmit chirp generated directly on the RF frequency (using a Digital Chirp Generator) before being amplified to the transmit level by a class C amplifier.

The receiver is even more essential, providing direct amplification of the received signal (with programmable gain and

band filtering) to an A-to-D converter operated in downsampling mode by digitising the signal at 26.6 MHz rate. In this way, the complete 10 MHz signal bandwidth can be represented unambiguously with only an acceptable amount of oversampling (30%) minimising the required hardware.

Instrument control and processing tasks are performed by the same AD-21020 DSP (with the help of a couple of FPGAs). The presumming can be varied from 1 to 32 in powers of two steps, and the resolution of science data can be selected to be 4, 6 or 8 bits, to allow optimisation of the data rate vs the operating scenarios.

The receive window position can either be controlled in open loop, using an a priori knowledge of S/C orbit and surface topography (which demonstrated to be a very robust approach) or in closed loop

Index Terms—penetrating radar, radar sounder.

I. MISSION OBJECTIVES AND CONSTRAINTS

Identification of water deposits, in either liquid or solid form, on or below the surface of Mars, is one of the current highest priorities in the exploration of the Red Planet. In particular, subsurface water deposits are suspected to exist as a relic of the ancient mars oceans.

Low-frequency radars, with their capability of penetrating deep into the planet surface, are natural candidates to the task of finding such deposits.

The MARSIS radar [1], operational onboard the ESA's Mars Express probe, is currently engaged in this mission.

While capable of great penetration (up to 5 km) the limited bandwidth of the system, imposed by the operational frequency, limits the range resolution and, consequently, its ability to discriminate radar returns close to the surface.

To complement the MARSIS investigation with an instrument capable of better resolution and ability to discriminate objects closer to the surface (at the expense of the penetration depth) the SHARAD (Mars Shallow Radar Sounder) instrument has been developed [2].

IV. ARCHITECTURE AND OPERATION

The SHARAD instrument operates on a carrier frequency of 20 MHz, and uses chirped pulses with a bandwidth of 10 MHz to achieve a nominal range resolution (in free space and without weighting) of 15 m. In order to allow detectability of weak subsurface echoes in presence of a strong surface return, very low range compression sidelobes are required (< -55 dBc after the 6th sidelobe) [3]. The antenna is a foldable dipole 10 metres long.

Synthetic aperture is used on-ground to achieve an along-track resolution down to 300 m in order to limit the unwanted echoes from off-nadir scatterers.

SHARAD has been designed to be installed on the "open-frame" MRO SpaceCraft, providing its own thermal control, minimising the use of S/C resources such as mass, power.

Another important driver of the SHARAD design, imposed by the challenging development schedule, was minimisation of the instrument hardware complexity and on-board processing tasks. The large downlink capability of the MRO S/C, in fact, fits both the desire of the designers to keep the processing simple, and that of the scientists to have raw or quasi-raw data on ground for deeper analysis.

III. INSTRUMENT MAIN ENGINEERING PARAMETERS

The SHARAD main engineering parameters are reported in Table 1. Note that in the nominal science altitude range, only the nominal PRF is used.

TABLE 1
MAIN INSTRUMENT PARAMETERS

Nominal science orbit altitude:	155-320 Km	
Extended/contingency orbit altitude:	230-407 Km	
Topographic margin:	-20/+10 Km	
Centre frequency:	20 MHz	
Chirp bandwidth:	10 MHz	
Pulse width:	85 usec	
Receive window width:	135 usec	
PRF (nominal):	700.28 Hz	Can work with halved PRF
PRF (low orbit):	775.19 Hz	Can work with halved PRF
PRF (high orbit):	670.22 Hz	Can work with halved PRF
Radiated power:	10 W	
Mass		
SEB (Electronics Box):	11.6 kg	
Antenna + cabling:	5.5 kg	
Power consumption		
Acquisition:	28W	
STD-BY:	13W	

A generic block diagram of the Instrument is provided in Figure 1. It can be seen that, basically, SHARAD operates in a very straightforward manner: the transmit chirp is synthesised directly at the transmit frequency (15 to 25 MHz) so that no conversion is needed, and amplified at the required power level before being sent to the matching network and the antenna, while the receive side uses a direct conversion approach with downsampling.

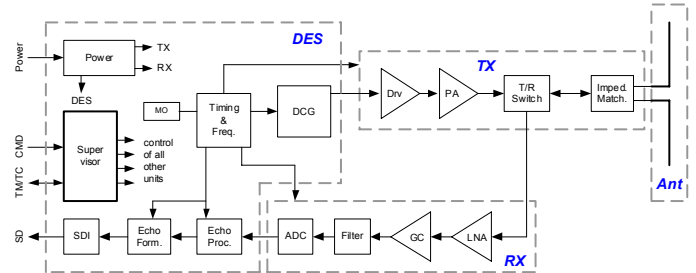


Figure 1 Block diagram of the SHARAD instrument.

Physically, the instrument is divided between the antenna (a 10 mt foldable dipole) and the SEB (SHARAD Electronic Box) which, in turn, is divided into two main blocks (see Figure 2):

- the RDS (Receiver and Digital Assembly), comprehensive of the controller and DSP functions and of the chirp generator (included in the DES – Digital Electronic Subsystem), and of the Rx module, and
- the TFE (Transmitter and Front-End) in charge of power amplification, antenna matching and Tx/Rx duplexing.

RDS and TFE are, physically, two separate boxes, mounted inside the mechanical structure of the SEB, which is a table-shaped structure, sizing 450 x 370 mm and 190 mm high, whose external side acts as a radiator for the thermal control, with the two boxes mounted “upside down” on the inner side. Both RDS and TFE have their internal DC/DC converters, and are powered directly by the S/C unregulated 28V power bus. The Antenna is a dipole, each arm 5 m long, consisting of two fiber tubes, used as a mechanical support structure for a metal wire which runs inside them and which represent the real “active” part of the antenna.

These two tubes, manufactured by ASTRO Aerospace (US) using the FFT (Fiber Foldable Tube) technology, are simply folded into 5 segments each, without using hinges/springs or other mechanical parts for the deployment, but relying only on the material elasticity to extend into the deployed position once released.

During the cruise phase, the antenna is kept stowed by two hinges, and enclosed in a Ge-kapton shield to protect it from the thermal stresses induced by the aerobraking.

The folded antenna has an envelope of 1524 x 241 x 190 mm.

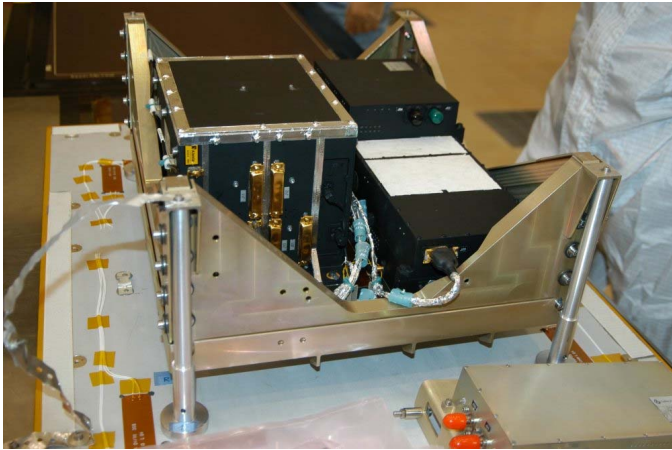


Figure 2 The SHARAD Electronics Box (SEB)

Coming to the functionalities of the electronics, the core of the chirp generator is a specialised ASIC which is capable to generate ‘continuous’ (vs the ‘stepped’ chirp achievable using common NCOs) chirps by incrementing the phase increment register on each 80 MHz clock step, with full programmability of slope and pulse duration.

The chirp generator also implements the Doppler compensation function (to compensate for the S/C radial velocity). Doppler compensation is implemented by introducing a phase shift between transmitted pulses, exploiting the programmability of the phase accumulator to add a phase offset, therefore compensating the Doppler shift (at centre frequency) by introducing a frequency shift on the transmitted signal. In the TFE, the signal to be transmitted is amplified to high power level before being radiated by antenna located close to the TFE unit.

The power amplification is provided by a two-stages BJT amplifier, providing an output power of 25 W peak (1 W peak the first stage), and operated at a regulated 28V collector voltage. The final stage uses a push-pull configuration in class C. The output of the final stage is matched on 50 ohm, and it is followed by an high-power PIN-diode switch which acts as a duplexer, and then by the antenna matching network, shared between the Tx and Rx paths.

The matching network shall achieve the maximum power transfer to/from the antenna which, in addition of being a balanced load, has an impedance which is rapidly changing with frequency (the natural frequency of the antenna is around 15 MHz, i.e. at the lower edge of the operation bandwidth). This is achieved using a 3-cells low-pass (LC) network, followed by a 4:1 transformer used also as balanced-unbalanced adapter (balun). This allows to achieve a return loss better 6 dB (75% power transfer efficiency) over the whole 15-25 MHz bandwidth.

Additionally, to make the transmitter more robust versus mismatched loads (in order to allow the instrument to operate – even if with degraded performances – also in case of improper or partial deployment of the antenna) a 2 db attenuator is added between the final stage and the switch.

The receiver side also uses minimal hardware, providing direct amplification of the received signal followed by analogue-to-digital conversion directly on the carrier.

At the input of the receiver chain there is a monolithic SPDT switch, which has the purpose of avoiding that the receiver chain is driven into saturation by the transmitted pulse leaking via the duplexer in the TFE.

Downstream the switch, after a first amplifier stage (made, as all the amplifiers within the Rx, using standard drop-in hybrid modules), there is the anti-aliasing filtering section.

To achieve the required band-shape over the 50% fractional bandwidth of the signal, this is not made with a band-pass filter: a cascade of high and low-pass filters is used, instead, with another gain stage in between to provide isolation.

Both filters are of Chebychev type, the high-pass of the 7th order and the low-pass of the 11th order.

This section is followed by a thermal compensation network (a PIN-diode attenuator driven by a temperature sensitive network, to minimise Rx chain gain variations over temperature).

The gain control function is implemented by two identical gain/attenuation stages, each with a 4 bit digital attenuator, in order to allow control of the Rx gain over a 30 dB range (the gain setting is programmed under control of the processor). Other two amplifier stages drive the A-to-D converter (AD 9042) which perform the conversion (with downsampling)

The received signal is downsampled at a frequency of 26.66 MHz (1/3 of the 80 MHz main reference frequency). Therefore, for the effect of the downsampling, the signal spectrum of 15-25 MHz is reversed and aliased to the range 1.66 – 11.66 MHz, with an oversampling factor of 33%.

It must be noted that, in the case of the SHARAD receiver, the noise figure is not the key element of the design, being the dominating contribution to the S/N ratio coming from the galactic background noise (estimated about 20 dB above kT).

After A-to-D conversion, the only processing applied to the received signal is a coherent presumming (performed in hardware by a dedicated FPGA) with variable presumming factors: 1 (no presumming), 2, 4, 8, 16, 28 and 32, to reduce the output data rate (see Table 2). Presumming is programmed (together with the number of bits to be transmitted) according to the operating scenario to achieve the minimum data rate compatible with the required data quality. In fact, being the focusing of the synthetic antenna performed on ground on the presumed data, the effect of the presumming is a “discretisation” of the applied phase law (same for all presumed samples), which lead to an increase of the synthetic antenna sidelobes proportional to the amount of the presumming. The resulting impact on data quality is dependent on the topography of the overflow region.

Phase coherence during presumming is maintained in presence of radial velocity components thanks to the Doppler compensation in the chirp generator. Full compensation is achieved only at centre frequency, but the residual errors at band edges are not such to introduce significant losses anyway.

The presumed samples are then packetised with 4, 6 or 8 bit resolution, using either a fixed or adaptive scaling (to better exploit the available dynamic range), by another FPGA, under the control of the ADSP 21020 instrument processor, and transmitted to the S/C data recorder.

The same ADSP 21020 is also in charge of the instrument commanding and monitoring, and all the other “intelligent” tasks within the instrument.

TABLE 2
INSTRUMENT DATA RATES VS PRESUMING/DATA WORD LENGTH

	data rate with header@nominal PRF (Mbit/sec)		
	BITS		
presuming	8	6	4
1	20.37	15.27	10.18
2	10.18	7.64	5.09
4	5.09	3.82	2.55
8	2.55	1.91	1.27
16	1.27	0.85	0.601
28	0.526	0.394	0.263
32	0.601	0.451	0.301

In the nominal conditions, the instrument operates with a PRF of 700.28 Hz, with nadir echo received in the next PRI (rank 1). It is possible to select, by command, an halved PRF (bringing SHARAD to operate in rank 0) in order to avoid possible ambiguities from returns from 40-50 degrees angle which may fall into rank 2 range (these returns are not significant in normal conditions, but could cause increase of clutter over particular topographies).

The nominal PRF of 700.28 Hz allows to cover the whole orbit range foreseen for the primary science phase, plus topographic margin. In order to allow operation over the wider orbit range required for MRO extended operation phase, two additional PRF (775.19 Hz and 670.22 Hz) can be selected (for both is also available the relevant halved PRF).

All instrument timings are derived from the same 80 MHz local oscillator, using an FPGA based state-machine.

SHARAD timing is reported in Figure 3.

The width of the receive window is 135 usec: 85 usec are allocated for the pulse width, 25 usec for the subsurface penetration (equivalent to 3.75 km in free space), 20 usec for the range window positioning margin, and 5 usec as margin for the ionospheric delay. Nominally, the surface echo is positioned 10 usec after the start of the receive gate.

The position of the receive window can be controlled either by *a priori* knowledge of the SHARAD altitude and surface topography (baseline operating mode), or by a closed-loop tracker (included as backup).

In the former case, the instrument will compute the position of the range window from:

- the orbital data (radius, radial velocity and latitude vs time) provided, before each observation, in a file (generated from the mission Flight Engineering

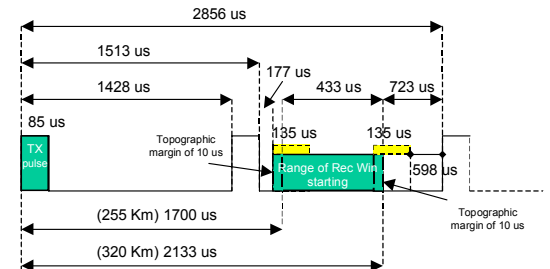
Team) called Orbit Data Table (ODT).

- the topographic profiles of the surface (expressed as a sequence of polynomials, each fitting a portion of the target surface corresponding to an acquisition time of, typical, 30 seconds), provided as parameters loaded into the SHARAD Parameter Table (PT) before each observation, as a function of the latitude.

The need for separate handling for orbit and surface data originates from the different level of planning at which they are produced: SHARAD planning (at least two weeks ahead) and Flight Engineering, being the orbit data updated no more than 1 week ahead of the passage to provide the required prediction accuracy.

In the closed-loop mode, SHARAD is able to track the surface using either a centre-of-gravity estimator or a threshold detector on the leading edge. In both case, tracking initialisation is performed using the open-loop data mentioned above. The ODT contains also radial velocity information used to define the Doppler compensation.

PRF: 700.28 Hz



PRF: 350.14 Hz

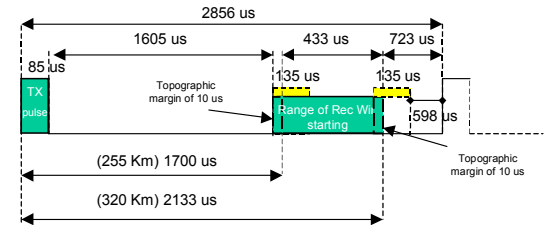


Figure 3 SHARAD instrument general timing.

V. INSTRUMENT MODES AND OPERATION SEQUENCE

SHARAD operates according to different operation and support modes, mainly:

- Std-by – when SHARAD is in StdBy only the RDS is powered. Receiver and TFE are switched OFF. The instrument is able to receive commands from the S/C. Std-By is the nominal mode while the instrument is not acquiring data.
- Warm-up 1 – the TFE is switched ON to warm up before an acquisition
- Warm-up 2 – like Warm-up 2 but also the receiver is ON
- SS (Subsurface Sounding) – this is the nominal

acquisition mode

- Rx-only – identical to SS but the transmission is not enabled: SHARAD operates as a receiver.
- Wait – transmission and acquisition of echoes is suspended

SHARAD operates in a pre-programmed way: when in STD-BY mode three separated files are loaded in the DES by the Spacecraft using dedicated commands:

- the OST (Operation Sequence Table), which contains the sequence of modes which shall be executed by the instrument, their duration, and mode parameters (like presuming, number of bits, receiver gain, etc)
- the ODT (Orbit Data Table) with the orbital data
- the PT (Parameter Table), with the target surface profiles

Following the “Enable” command from the S/C, the instrument will then start its operational sequence according to the OST, ODT, PT, to return to the Std-By mode at the end of the sequence. No commands from the S/C are accepted during the execution of a sequence.

VI. INSTRUMENT PERFORMANCE

Very low range sidelobes are mandatory to allow detection of weak sub-surface echoes in presence of the strong surface return.

The approach of limiting to the minimum the amount of processing performed on-board was selected in order to allow the pulse compression process (critical from the point of view of range sidelobes) to be performed on ground, using the computed instrument Point Target Response as correlation reference waveform [4]. This is derived from:

- the electronics PTR, measured during instrument test
- the antenna response: this is computed from the response of a flat surface during the calibration phase.

Due to the above approach, the critical parameter is the stability of the Tx/Rx distortions instead of their absolute value. During thermal-vacuum tests on the SEB, the reference function acquired in ambient conditions has been used to correlate simulated echoes collected over the whole temperature range: the achieved (weighted) Point Target responses, (showed in Figure 4) were well within the required limits (-55 dBc for far lobes).

This excellent behaviour over temperature provided also an indirect feedback about the small sensitivity of the frequency response to component drifts, supporting the analytical verification, and the assumption that the on-ground characterisation data will be valid for good in-flight data correlation through the whole operating life of the instrument.

The effect of the antenna, which could not be characterised on-ground, has been derived by in-flight calibration, performed by passive acquisition of the galactic noiselevel vs frequency and by collection of echoes from flat surfaces. This has allowed to correct the SEB reference waveform to account also for the antenna response.

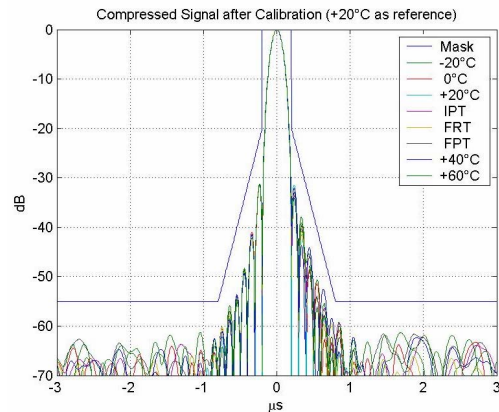


Figure 4 SHARAD range sidelobes after calibration vs temperature

VII. CONCLUSIONS

The SHARAD instrument provides significant performance and operational capability with a quite simple design. The use of direct synthesis on carrier of the chirp signal, and down-sampling directly from RF in the receive has allowed to keep the overall complexity quite low, with benefits on mass, power and reliability.

Successful operation of the instrument on Mars has demonstrated that the initial design goals have been met. Sharp range responses are still achieved after almost 3 years from the on-ground characterisation, demonstrating the stability of the hardware, and the instrument is performing nominally with respect to all other performances.

SHARAD is currently providing high-quality science data to the scientific community.

VIII. ACKNOWLEDGEMENTS

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